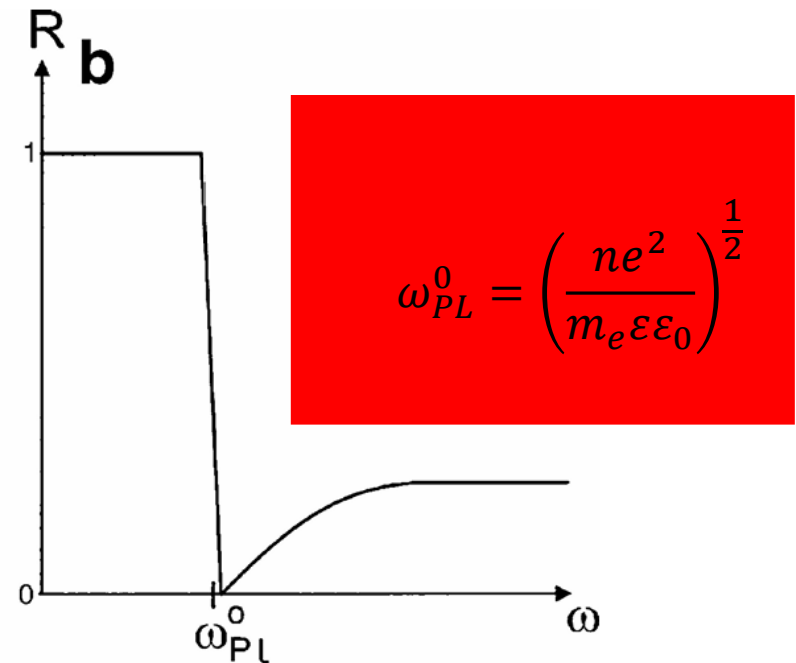
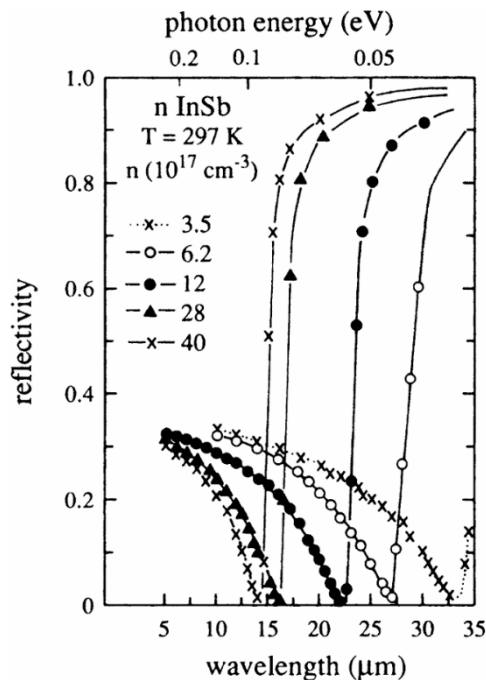


CHAPTER 12

Optical Properties of Plasmons, Plasmon-Phonon Mixed States and of Magnons

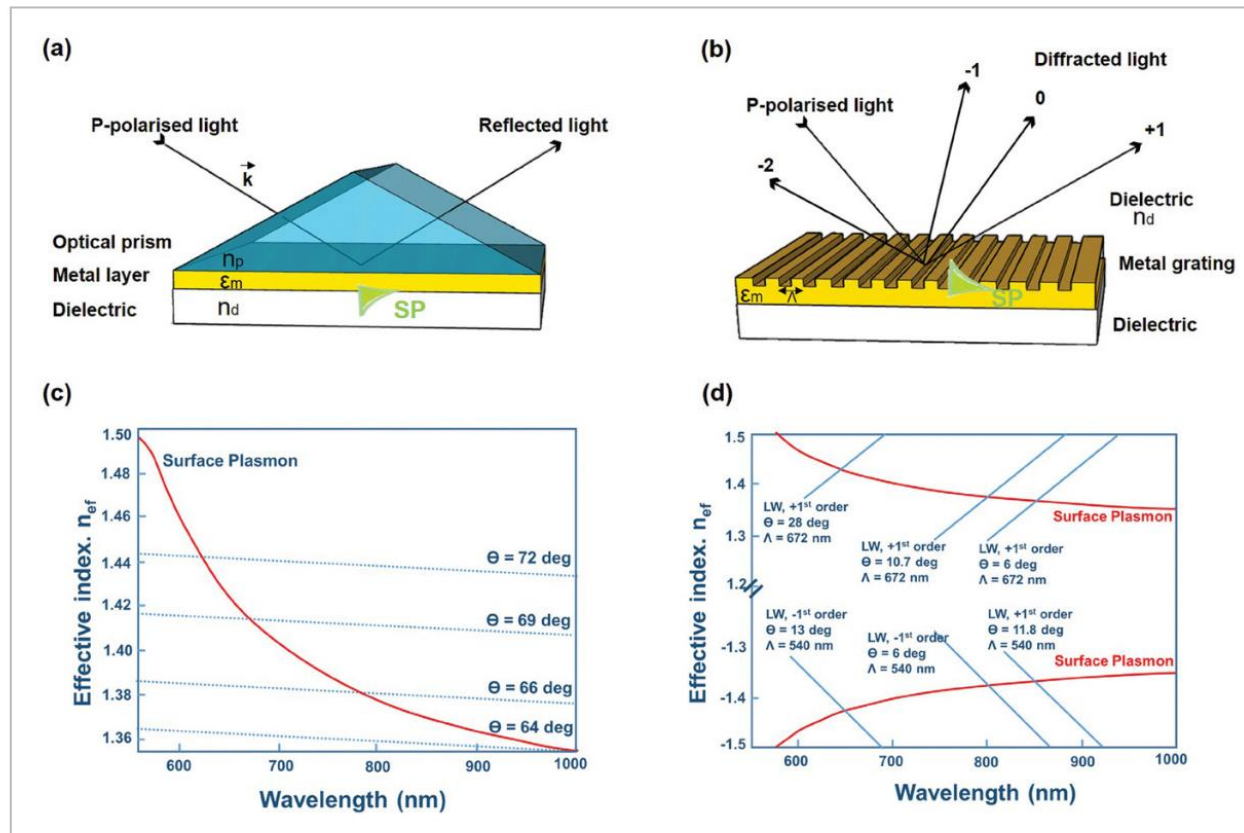
- Plasmons as collective excitations of the carriers in a partly filled band usually do not exist in intrinsic semiconductors under thermodynamic equilibrium
- Two conditions under which plasmons can be observed, high doping or high excitation
 - In highly doped semiconductors, there is a high density either of electrons or of holes
 - In highly excited semiconductors, an electron–hole plasma can be formed, which consists of electrons and holes



The reflection spectrum in the vicinity of the plasmon resonance in doped InSb samples

12.1 Surface Plasmons

- Surface plasmons are electron collective oscillations that exist at the interface between a metal and a dielectric material.
- The surface plasmons can be investigated either by attenuated total reflection or by a grating with spacing Λ engraved in the surface.



12.2 Plasmon-Phonon Mixed States

- When $\omega_{PL} \approx \omega_{LO}$ there are plasmon-phonon mixed states

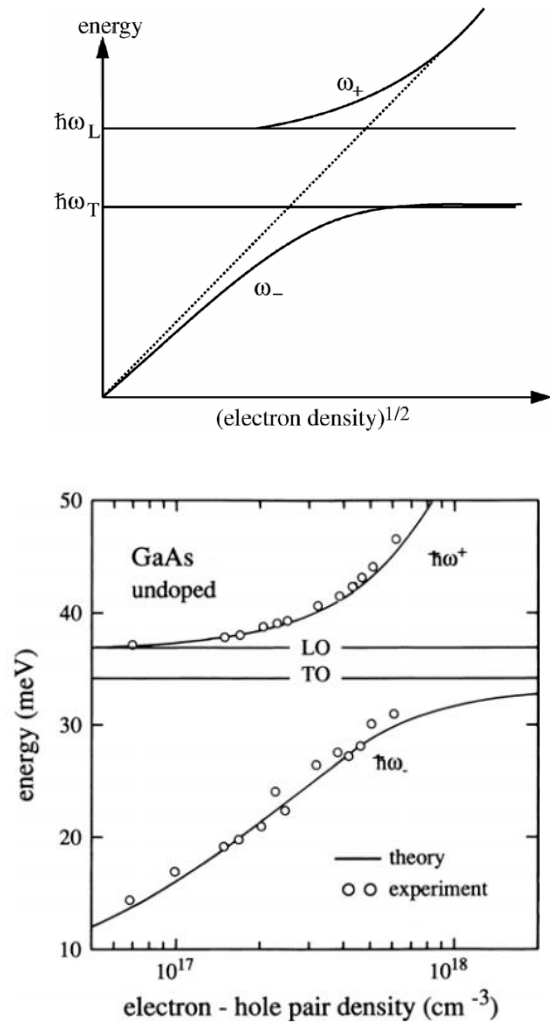


Fig. 12.3. The dependence of the plasmon-phonon mixed mode on the electron-hole pair density in highly photon-excited GaAs [84N1]

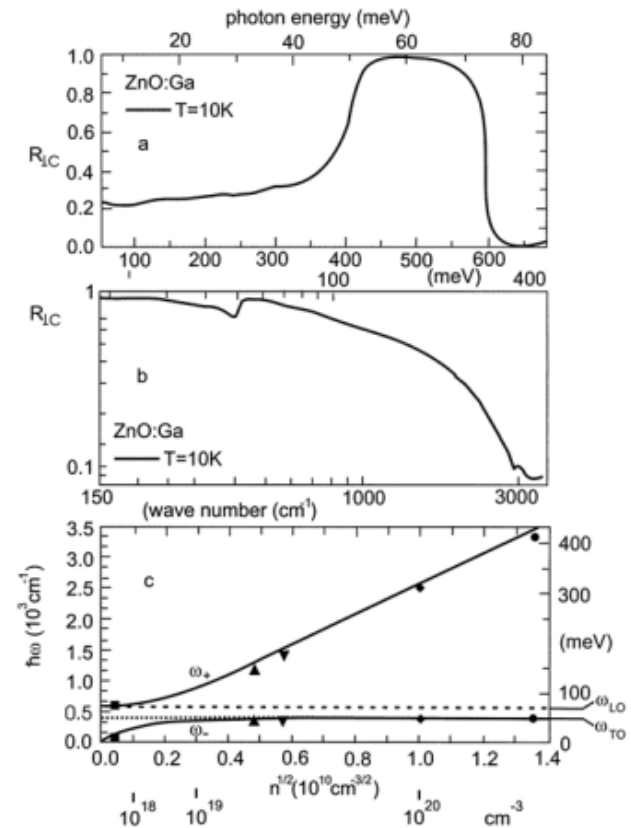


Fig. 12.4. Reflection spectra of n -doped ZnO:Ga samples with two different doping levels (note the different abscissas) (a,b) and the various transverse and longitudinal eigenenergies of the plasmon-phonon mixed states as a function of carrier density (c) [97G1]

12.3 Plasmons in Systems of Reduced Dimensionality

■ In Fig. 12.5 we show two different modulation-doped GaAs/Al_{1-y}Ga_yAs samples.

- The data points have been deduced from angle-resolved Raman scattering, which allows variation of k_p
- If the quantum wells couple in a MQW, also a transverse wavelength or $k_{\perp}$
- A dispersion for $\omega_{PL}(k_p, k_{\perp})$ taking such effects into account is given:

(12.4)

$$\omega_{PL} = \left\{ \frac{2\pi n e^2}{\epsilon_M m_{eff}^*} k_p \frac{\sin k_p d}{\cosh k_p d - \cos k_{\perp} d} \right\}^{1/2}$$

- The solid lines have been calculated with (12.4) and the dashed lines are linear approximations.

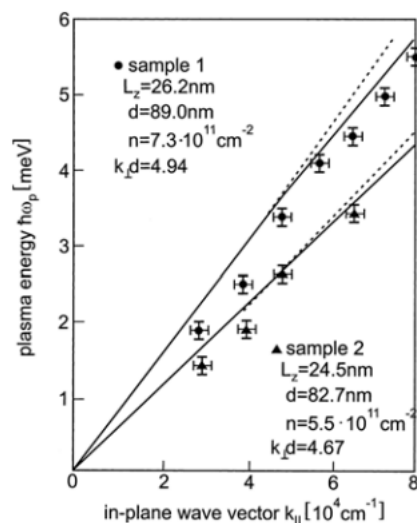


Fig. 12.5. The in-plane dispersion relation of plasmons in two different modulation doped GaAs/Al_{1-y}Ga_yAs samples [8201]